

DeLLMa: How It Works

DeLLMa chooses one action that maximizes *expected utility* under a *fixed* belief over uncertain states. The belief is elicited from an LLM (verbal labels $\rightarrow$ scores $\rightarrow$ normalize); utilities come from the LLM’s rankings of state–action pairs, aggregated via Bradley–Terry. There is no Bayesian update. Two examples: agriculture (crop choice) and stocks (which stock to buy).

1 Notation

- **Actions** $A = \{a_1, \dots, a_n\}$: the options (e.g., plant apples, buy AMD).
- **State** $\theta \in \Theta$: a vector of uncertain factors (e.g., climate, demand).
- **Context** C : all inputs (reports, tables, summaries).
- **Utility** $U(\theta, a)$: value of action a in state θ (higher is better).

2 Goal

DeLLMa picks the action that maximizes expected utility:

$$U_C(a) = \mathbb{E}_{\pi(\theta|C)}[U(\theta, a)] = \sum_{\theta \in \Theta} \pi(\theta | C) U(\theta, a), \quad (1)$$

$$a^* = \arg \max_{a \in A} U_C(a). \quad (2)$$

The distribution $\pi(\theta | C)$ is a **fixed belief** over states given context C : it is set once (elicited from the LLM or loaded from cache) and **not updated** during the pipeline.

3 Mathematical Process

3.1 State space

There are k factors (e.g., climate, supply chain, per-product price change). Each factor f_i has ℓ possible values (e.g., $\ell = 3$). A state is $\theta = (\theta_1, \dots, \theta_k)$ with $\theta_i \in \{\tilde{f}_i^1, \dots, \tilde{f}_i^\ell\}$; the full state space has size ℓ^k and is not enumerated—states are sampled.

3.2 Belief $\pi(\theta | C)$

For each factor and each value, the LLM gives a verbal label (e.g., “very likely”, “likely”). A fixed score map V converts labels to numbers:

very likely	6
likely	5
somewhat likely	4
somewhat unlikely	3
unlikely	2
very unlikely	1

Scores are normalized per factor, then multiplied across factors (independence):

$$\pi_i(\tilde{f}_i^j \mid C) = \frac{V(v_{ij})}{\sum_{j'=1}^{\ell} V(v_{ij'})}, \quad \pi(\theta \mid C) = \prod_{i=1}^k \pi_i(\theta_i \mid C). \quad (3)$$

So the belief is **elicitation** + **normalization**; there is no prior/likelihood or Bayesian update.

3.3 Sampling and state–action pairs

- Sample s states from $\pi(\theta \mid C)$ (for each factor, draw from its marginal).
- Form all pairs $(\theta^{(j)}, a_i)$ for $j = 1, \dots, s$ and each action a_i —so each state is paired with every action (variance reduction). Total: $N = s \cdot n$ pairs. Shuffle.
- Split into overlapping minibatches of size b ; stride between batch starts is $\lfloor b(1 - q) \rfloor$ with overlap q .

In words: we draw many possible “futures” $\theta^{(j)}$ from the belief, pair each future with *every* action so we can compare actions under the same conditions, and then group these pairs into small batches that the LLM can comfortably rank.

Index the N state–action pairs by $i = 1, \dots, N$. Let $\text{act}(i)$ be the action in pair i , and define

$$I_a = \{i \in \{1, \dots, N\} : \text{act}(i) = a\},$$

the set of indices of pairs that use action a .

3.4 Rankings to preferences

The LLM ranks the pairs in each minibatch (e.g., r_1 best, r_b worst). Two options:

- **Rank2Pairs**: add preference $r_i \succ r_j$ for every $i < j$.
- **Top1**: add only $r_1 \succ r_j$ for $j = 2, \dots, b$.

Call the set of pairwise comparisons

$$\Omega \subseteq \{1, \dots, N\} \times \{1, \dots, N\},$$

where $(i, j) \in \Omega$ means “pair i is preferred to pair j .”

In words: each minibatch ranking is turned into many simple “ i beats j ” facts. Collecting these across all minibatches gives a large dataset Ω of wins and losses between state–action pairs.

3.5 Bradley–Terry and scores

Each state–action pair i is treated as an *item* with a (latent) **strength** or **utility** parameter $w_i \in \mathbb{R}$. Intuitively, a larger w_i means “pair i is better” according to the comparisons. The Bradley–Terry model specifies the probability that item i is preferred to item j as

$$P(i \succ j) = \frac{e^{w_i}}{e^{w_i} + e^{w_j}}. \quad (4)$$

Given the observed comparisons Ω , the log-likelihood (with optional L_2 regularization) is

$$\mathcal{L}(\mathbf{w}) = \sum_{(i,j) \in \Omega} \log \frac{e^{w_i}}{e^{w_i} + e^{w_j}} - \lambda \sum_{i=1}^N w_i^2, \quad (5)$$

where $\lambda \geq 0$ controls the regularization strength. The strength vector $\mathbf{w} = (w_1, \dots, w_N)$ is fitted by maximizing $\mathcal{L}(\mathbf{w})$, using an iterative algorithm (iterative Luce–spectral ranking, ILSR). Optionally, a softmax with temperature τ is applied to obtain normalized scores $\tilde{w}_i$.

3.6 Action utility and decision

For each action a , average the (possibly softmaxed) scores over pairs that have that action:

$$\bar{U}(a) = \frac{1}{|I_a|} \sum_{i \in I_a} \tilde{w}_i. \quad (6)$$

Then

$$a^* = \arg \max_{a \in A} \bar{U}(a). \quad (7)$$

This approximates the expected utility in (1) by Monte Carlo over the sampled states; the Bradley–Terry scores play the role of $U(\theta, a)$.

3.7 Pipeline in one line

Elicit $\pi(\theta \mid C) \rightarrow$ sample states $\rightarrow$ form pairs $\rightarrow$ minibatches $\rightarrow$ LLM ranks $\rightarrow$ pairwise $\Omega \rightarrow$ ILSR $\rightarrow \mathbf{w} \rightarrow$ (optional softmax) $\rightarrow \bar{U}(a) \rightarrow a^* = \arg \max_a \bar{U}(a)$.

4 Data Used

DeLLMa uses only:

- **Context C :** reports, summaries, tables fed into the LLM.
- **Cached beliefs (optional):** a precomputed π stored in JSON, so the LLM is not queried again for the belief step.

There is no observation that triggers a Bayesian update.

5 Example 1: Agriculture

Context: USDA fruit report and California price/yield statistics for the chosen fruits.

States: climate, supply chain, per-fruit price and yield change (discretized).

Actions: e.g., “Plant 10 acres of apples”, “Plant 10 acres of avocados”, ...

Flow: C (and optionally cached beliefs) $\rightarrow \pi(\theta \mid C) \rightarrow$ sample $\rightarrow$ rank $\rightarrow$ Bradley–Terry $\rightarrow \bar{U}(a) \rightarrow a^*$.

6 Example 2: Stocks

Context: Historical monthly stock prices and current price for the chosen tickers; buy/sell dates.

States: economic health, market sentiment, per-stock factors (growth, launches, etc.).

Actions: e.g., “Buy AMD”, “Buy GOOGL”, ...

Flow: Same as above.

7 Extension: Power Grid (Estimating Inertia M in IEEE 14-Bus)

DeLLMa can be extended to *active probing* in power systems, for example to choose a small excitation that best helps estimate the inertia M of generators in the IEEE 14-bus test system.

Setup (IEEE 14-bus inertia estimation):

- **State θ :** unknown dynamic parameters and operating conditions, e.g., generator inertias M , damping D , loading level, and which lines or units are in/out of service.
- **Context C :** known grid information that is passed (in text or tables) to the LLM, for example:
 - IEEE 14-bus topology and nominal parameters (bus, line, generator data);
 - current operating point: generator outputs, loads, voltages, and interchange;
 - past disturbance–response records or simulation summaries (plots or descriptions of frequency, rotor angle, and power following small events);
 - operating limits and safety constraints for probes (allowed size and duration of perturbations);
 - a short description of the identification method (how M will be estimated from measured frequency).

In short, C is everything you already know about the IEEE 14-bus system and how you will process the response.

- **Actions A :** candidate *active probes* that DeLLMa will choose between, such as:
 - a_1 : 1% step increase in active power setpoint of a chosen generator for 2 seconds;
 - a_2 : 2% pulse on another generator;
 - a_3 : small oscillatory modulation of load at a selected bus;
 - etc.

Each $a \in A$ is assumed to be safe and within operational limits. The *decision problem* DeLLMa solves is: “Given the current IEEE 14-bus context C and uncertainty about M , which probe $a \in A$ should we apply to make our inertia estimate as accurate and reliable as possible?”

- **Utility $U(\theta, a)$:** quality of the inertia estimate after applying probe a in state θ , for example

$$U(\theta, a) = -\text{Var}(\widehat{M} \mid \theta, a),$$

so higher utility means a more informative probe (lower expected variance of the estimator $\widehat{M}$) given the same safety constraints.

How DeLLMa applies: using the general pipeline,

1. Use C to elicit a fixed belief $\pi(\theta \mid C)$ over the unknown dynamic state (plausible values of M , D , loading, topology).
2. Sample states $\theta^{(1)}, \dots, \theta^{(s)} \sim \pi(\theta \mid C)$ and form state–action pairs $(\theta^{(j)}, a_i)$ for all j and $a_i \in A$.
3. For each pair, describe to the LLM the operating condition and the probe (e.g., “IEEE 14-bus near base load, inertia likely in $[X, Y]$; apply probe a_i .”), and let the LLM rank the pairs in minibatches according to how well they will allow M to be estimated.
4. Convert rankings to pairwise comparisons Ω , fit Bradley–Terry to obtain strengths w_i for all pairs, and compute $\bar{U}(a)$ via (6).
5. Choose the probe

$$a^* = \arg \max_{a \in A} \bar{U}(a),$$

that is, the active probe expected (under $\pi(\theta \mid C)$) to give the most informative frequency response for estimating inertia M in the IEEE 14-bus system.

Caveats. (1) Probes must satisfy all operational constraints (voltage, line flows, frequency); only safe actions should be included in A , and they may be further screened by a power-flow or dynamic simulation. (2) The procedure is suited to *planning* experiments (e.g., off-line studies or scheduled tests), not real-time emergency control. (3) Quantitative evaluation requires a simulator or historical disturbance data to link the chosen probe to actual estimation accuracy for M .

8 DeLLMa vs. Bayesian Optimization

	DeLLMa	Bayesian optimization
Goal	One action a^* (max expected utility)	Sequence of points (max unknown f)
Belief	Over states θ ; fixed, not updated	Over objective f ; prior $\rightarrow$ posterior
Update	None	After each observation (x_i, y_i)

DeLLMa is **not** Bayesian optimization: it uses a fixed belief and makes one decision; BO updates a prior to a posterior and designs a sequence of experiments.

9 Summary

DeLLMa maximizes expected utility under a **fixed** belief $\pi(\theta \mid C)$ elicited from an LLM. Utilities are inferred from the LLM’s rankings via Bradley–Terry. No Bayesian update; the only “data” is the context C (and optionally cached beliefs). It is a decision-under-uncertainty procedure, not sequential optimization like BO.

Liu et al. (2024), DeLLMa. arXiv:2402.02392.